

## Tiled Chip Multicore Processors



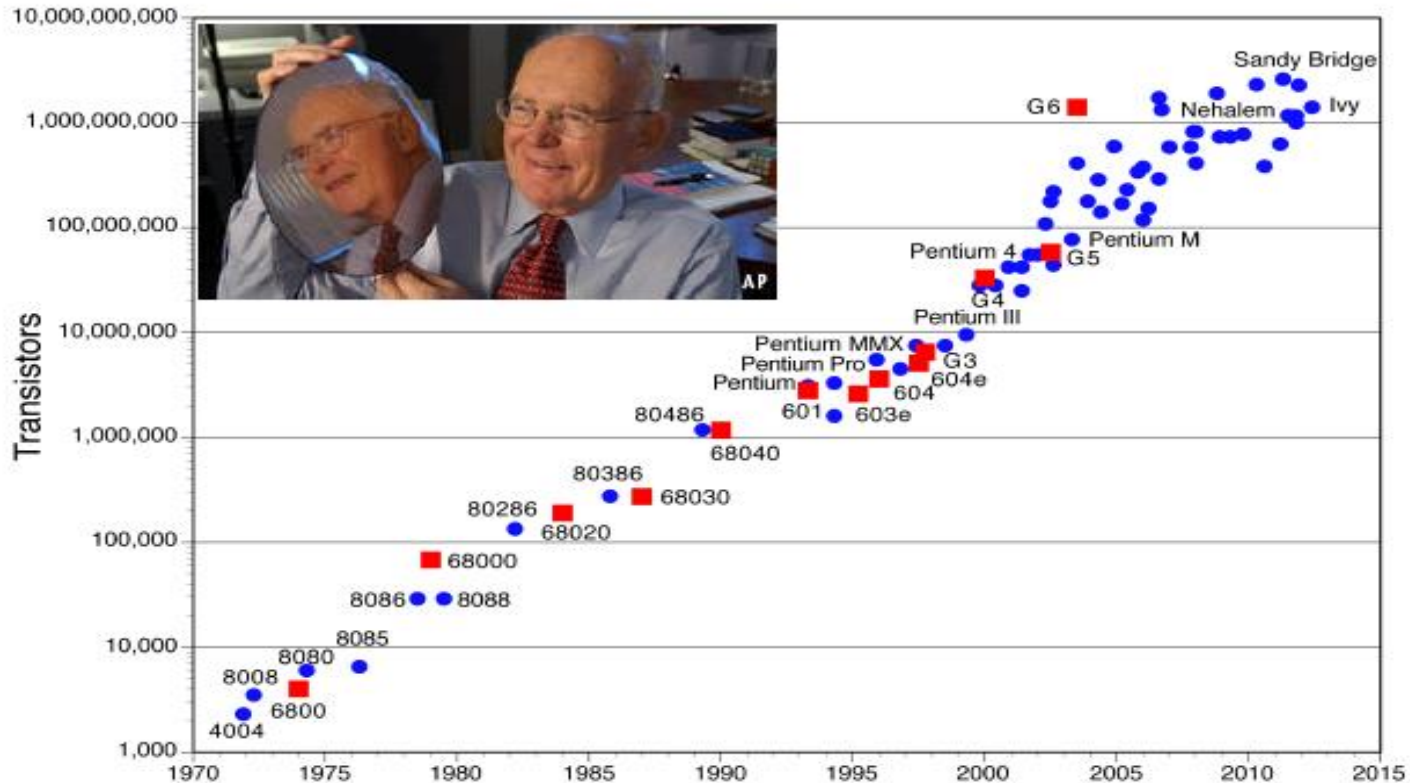
**John Jose**

**Associate Professor**

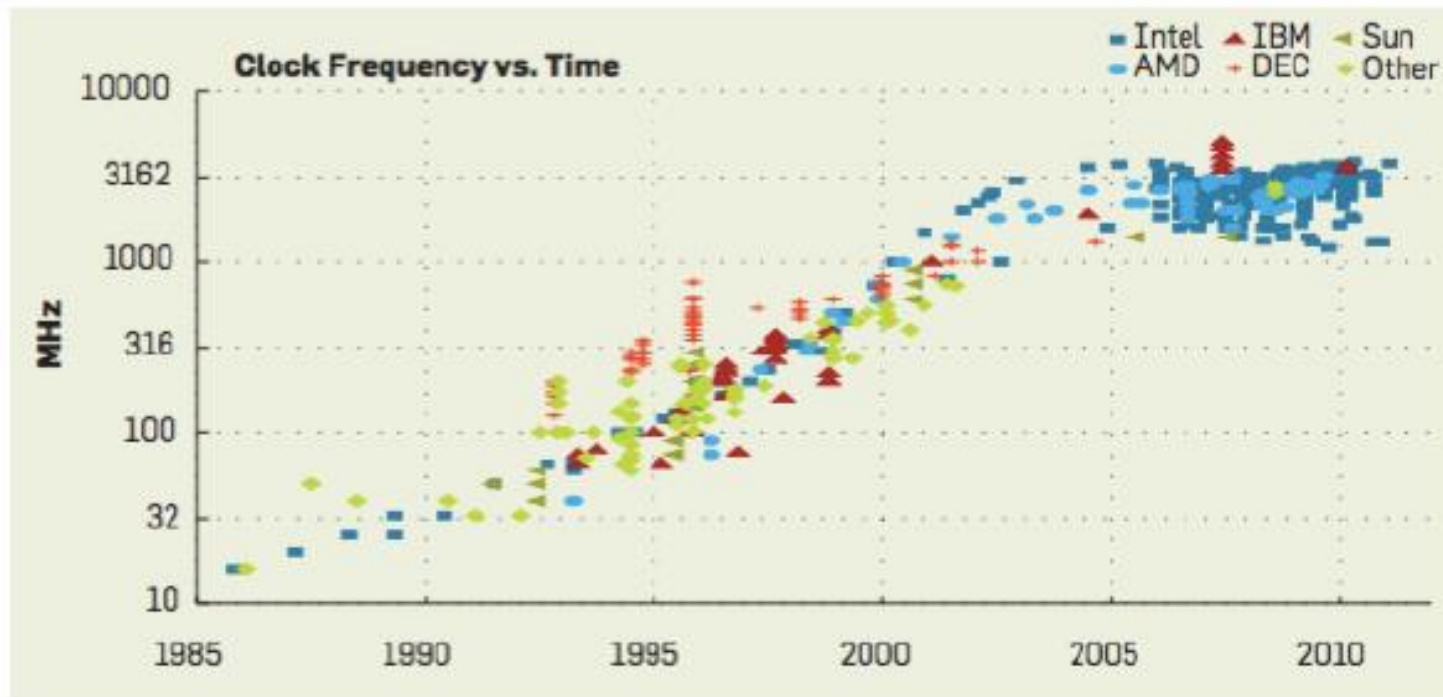
**Department of Computer Science & Engineering**

**Indian Institute of Technology Guwahati**

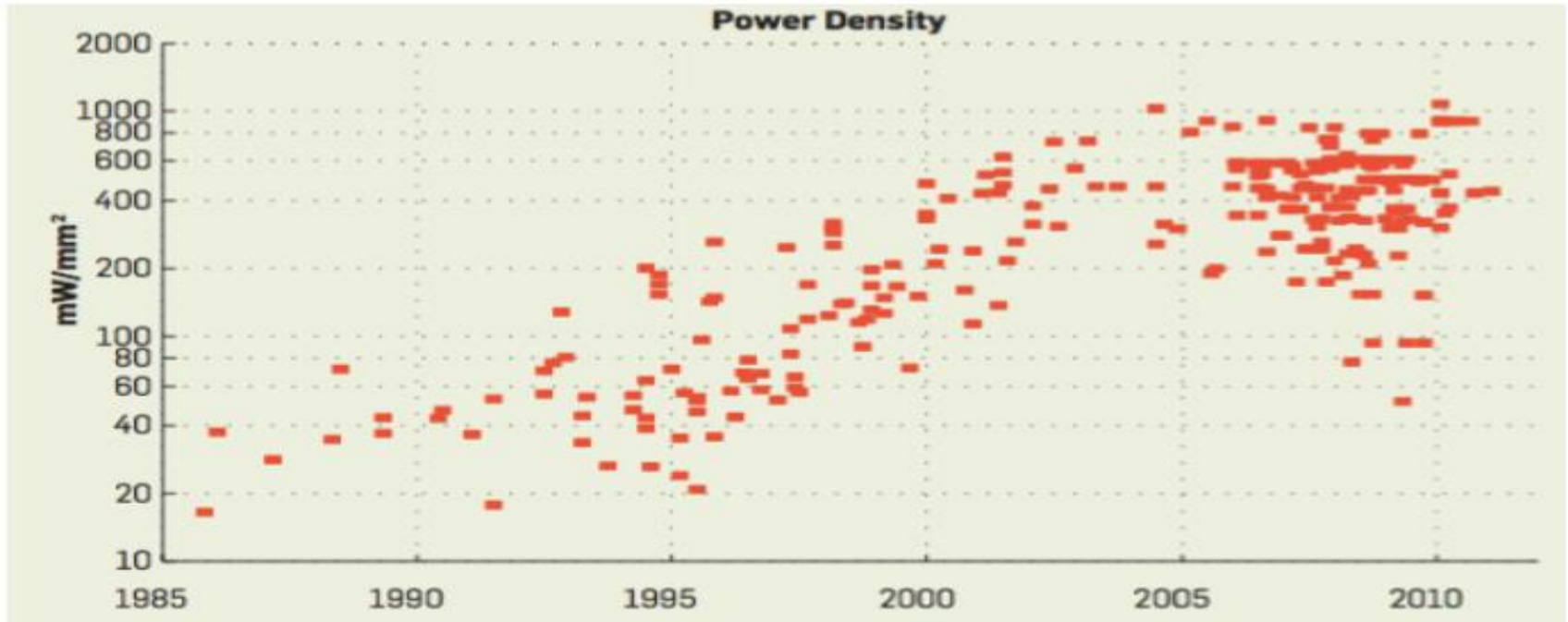
# Impact of VLSI in processor trends



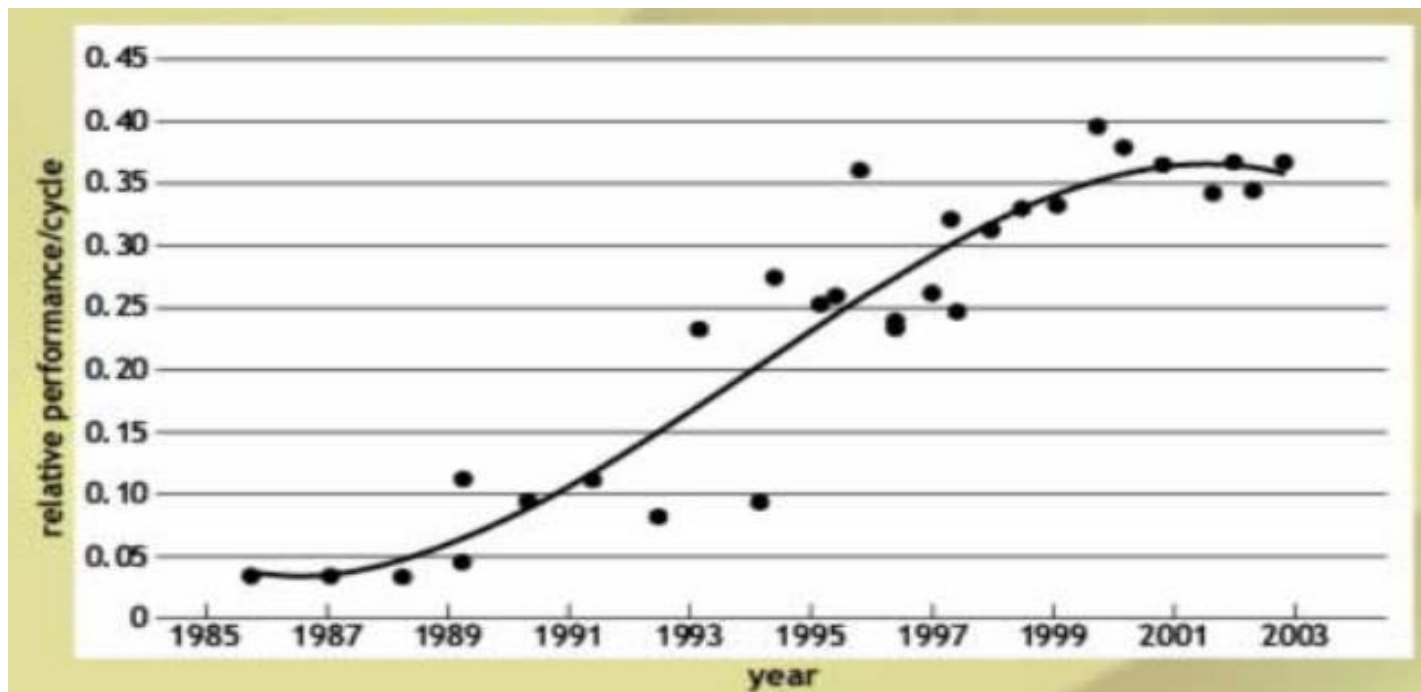
# Frequency Wall



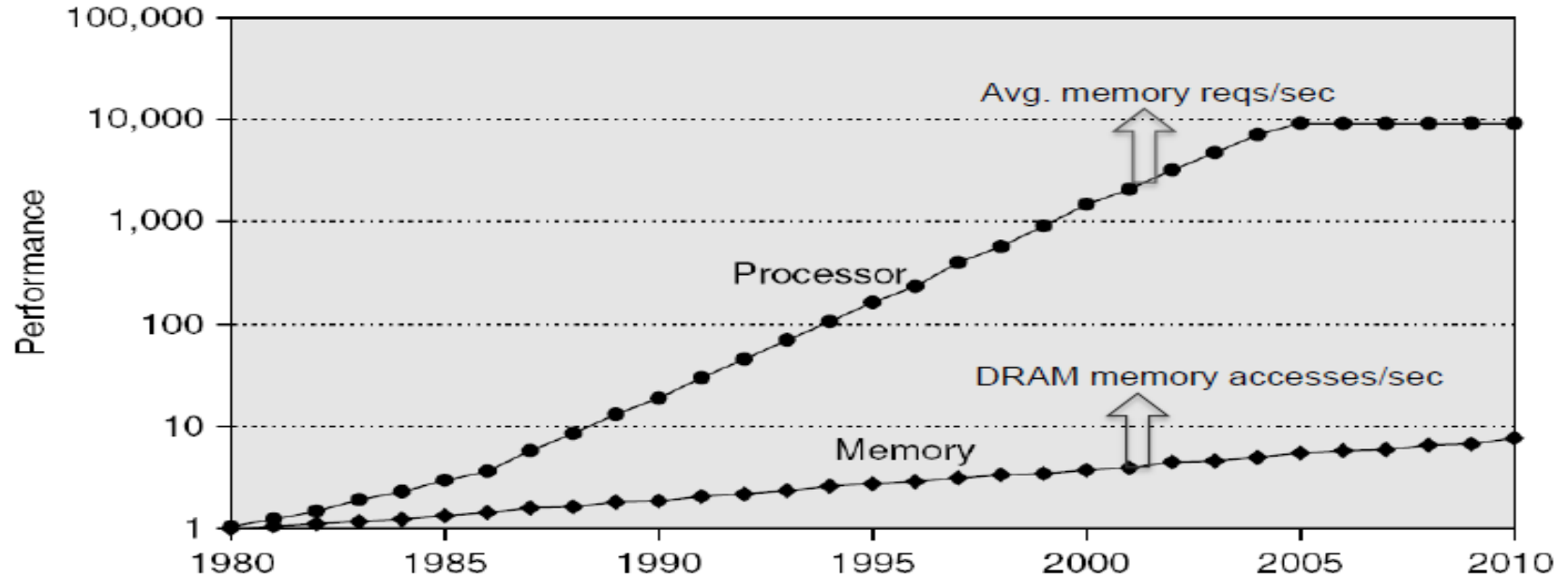
# Power Wall



# ILP Wall

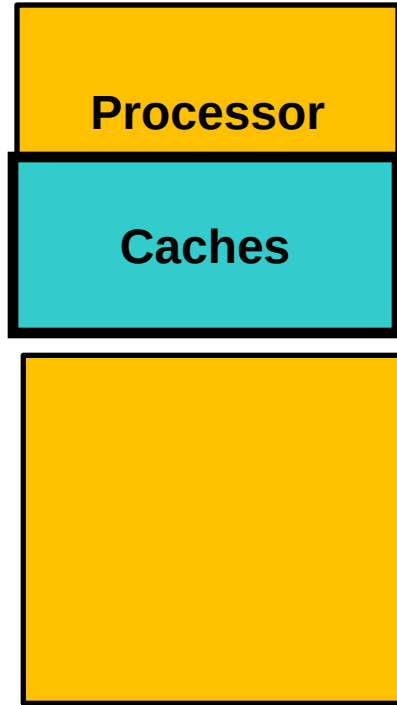


# Memory Wall



# Paradigm Shift to Multicore

2004 SoC : 90 nm

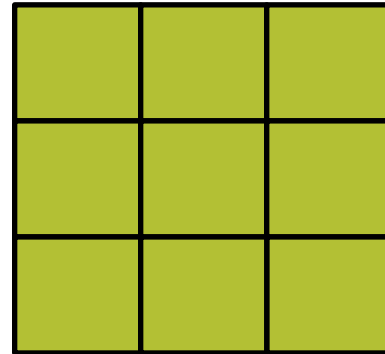
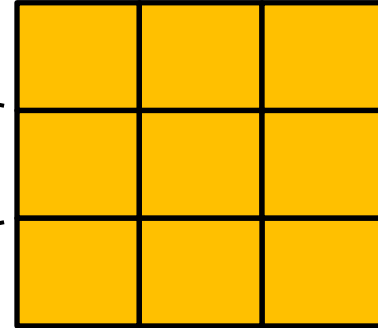


P  
O  
W  
E  
R

P  
E  
R  
F



2017 SoC : 10 nm



P  
O  
W  
E  
R

P  
E  
R  
F

Multiple slower processors vs single fast powerful processor

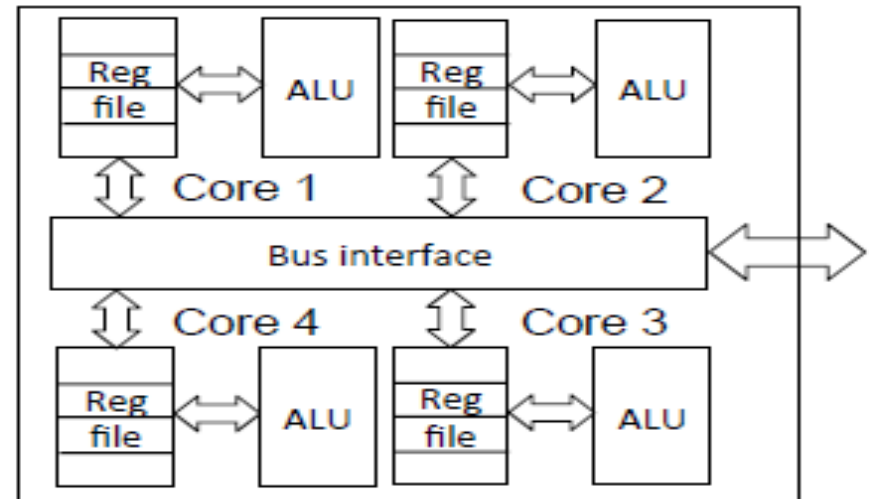
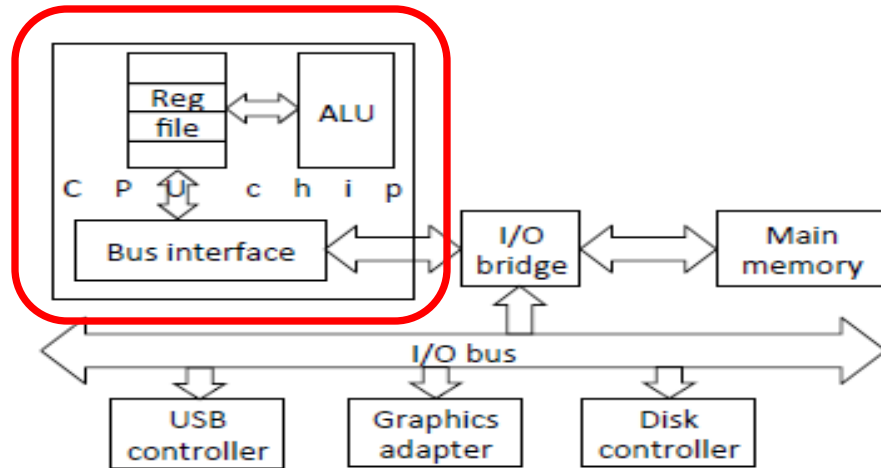
# Paradigm Shift to Multicore



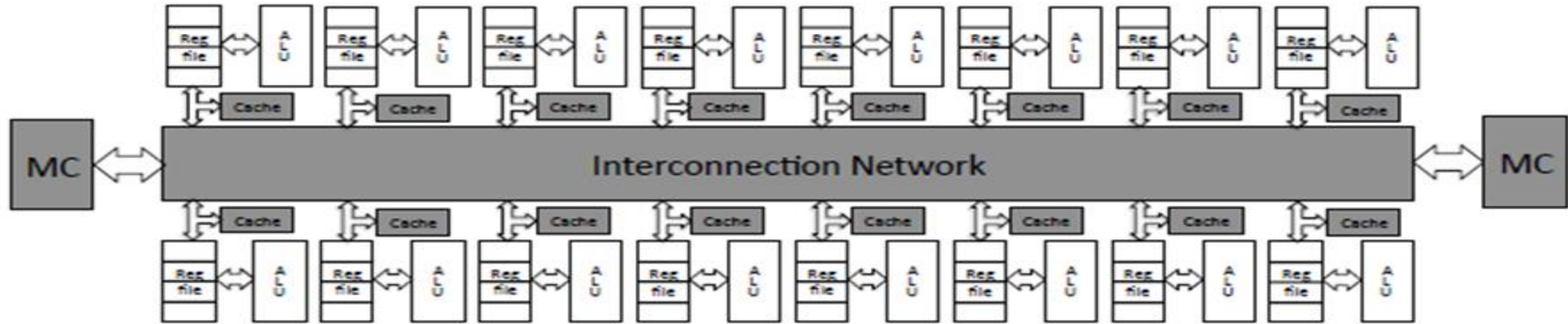
**Multiple slower processors is better than single fast powerful processor**



# What is Multicore ?



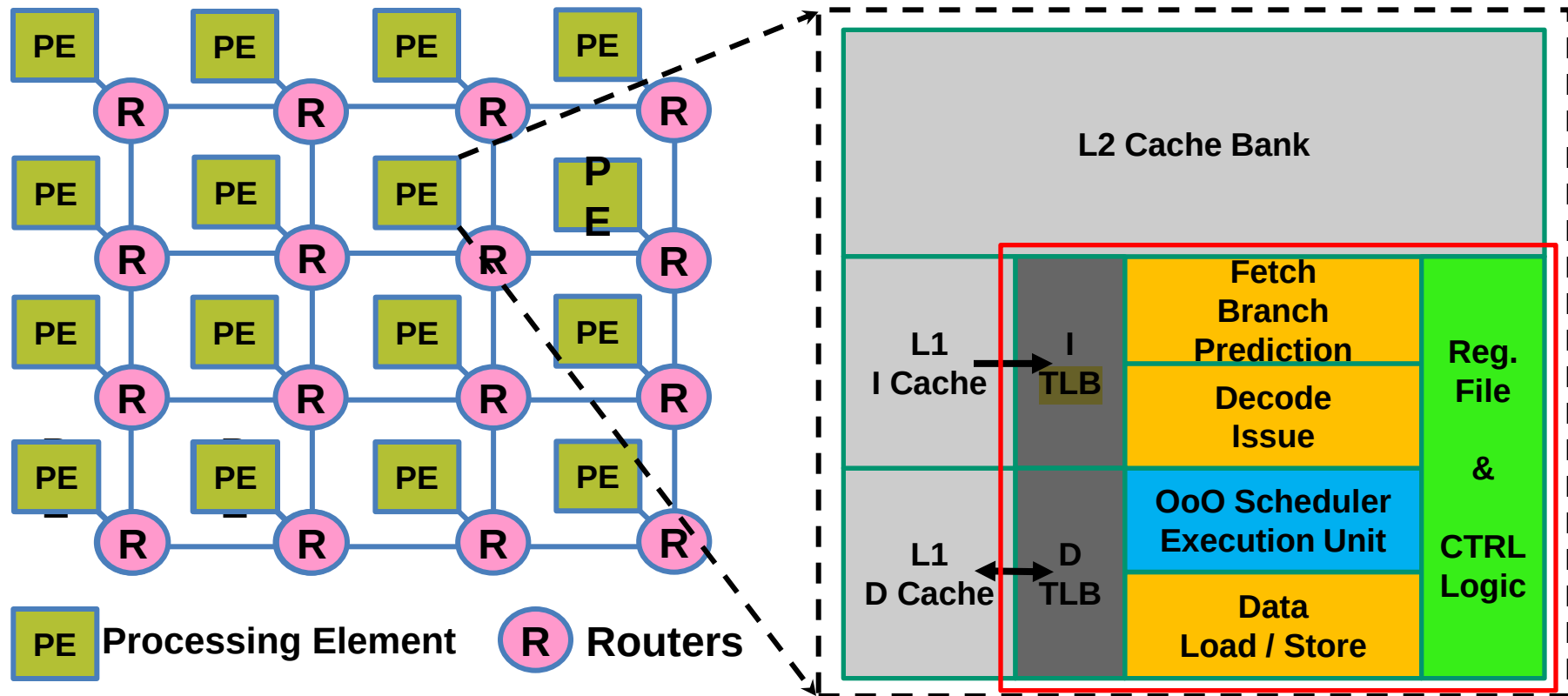
# What is Multicore ?



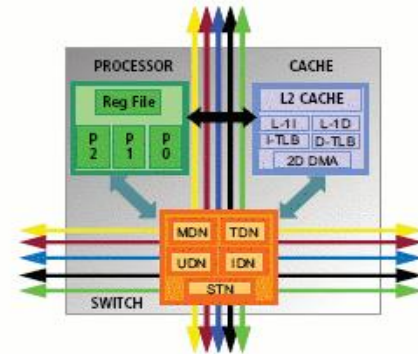
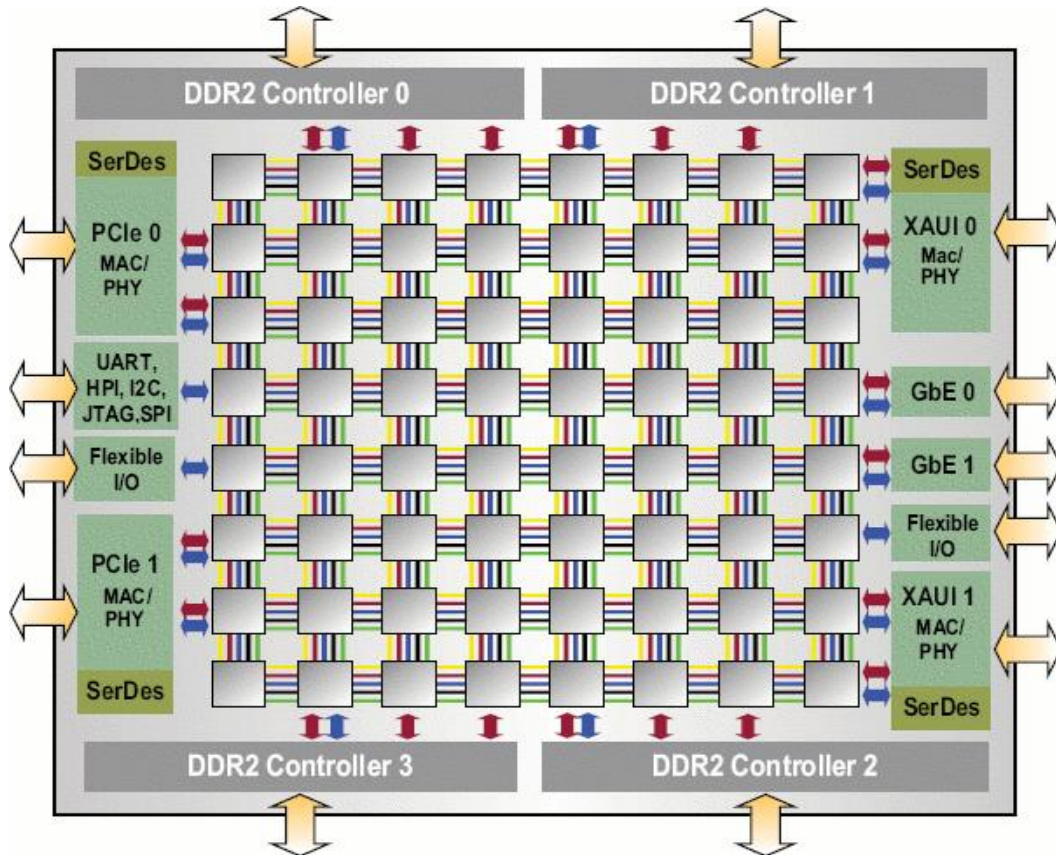
❖ How these cores communicate ?

❖ What is the best interconnection mechanism ?

# Tiled Chip Many-Core Processor (TCMP)

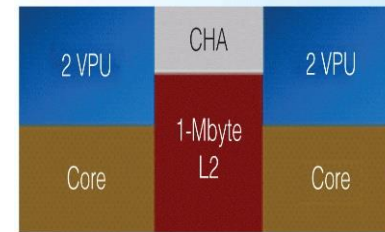
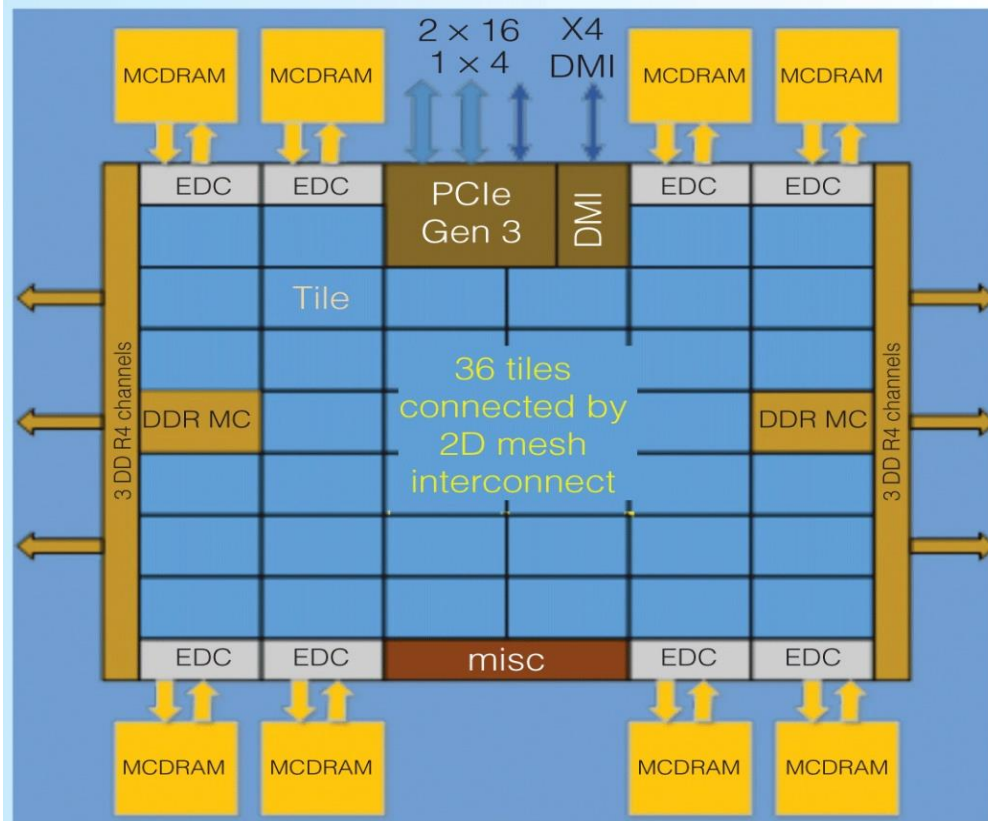


# State-of-the-Art Architectures



Tilera64

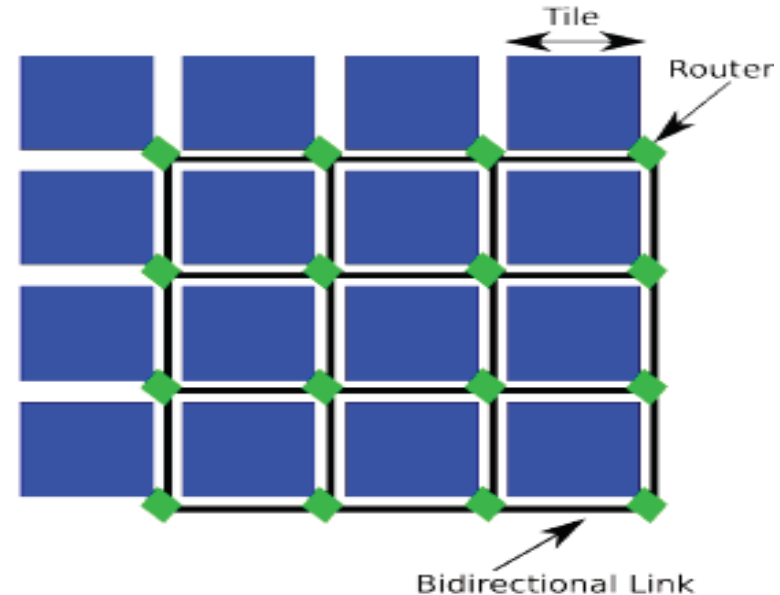
# State-of-the-Art Architectures



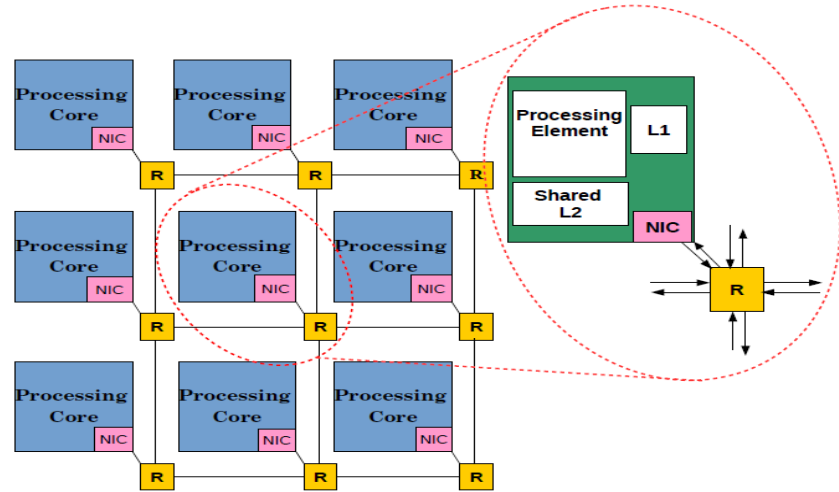
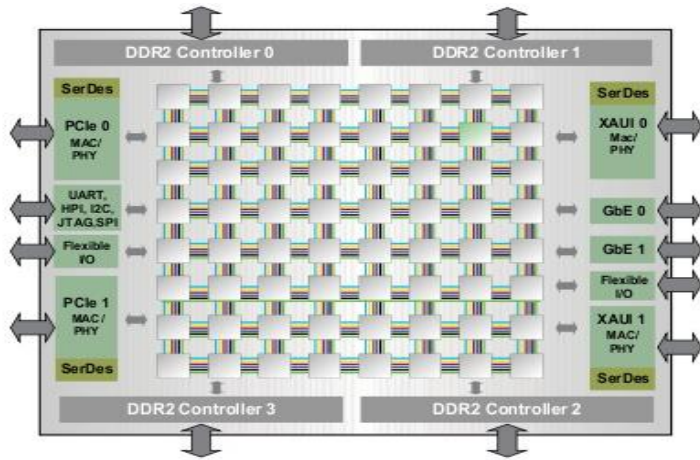
**Intel KNL**

# What is NoC ?

- ❖ Processing units interconnected via packet based network
- ❖ Each resource is called as a ***'tile'***
- ❖ All resources organized as rectangular tiles on the chip.
- ❖ Each tile have an address -  $(X, Y)$
- ❖ Tiles interconnected by network of routers
- ❖ Communication by packet transmission

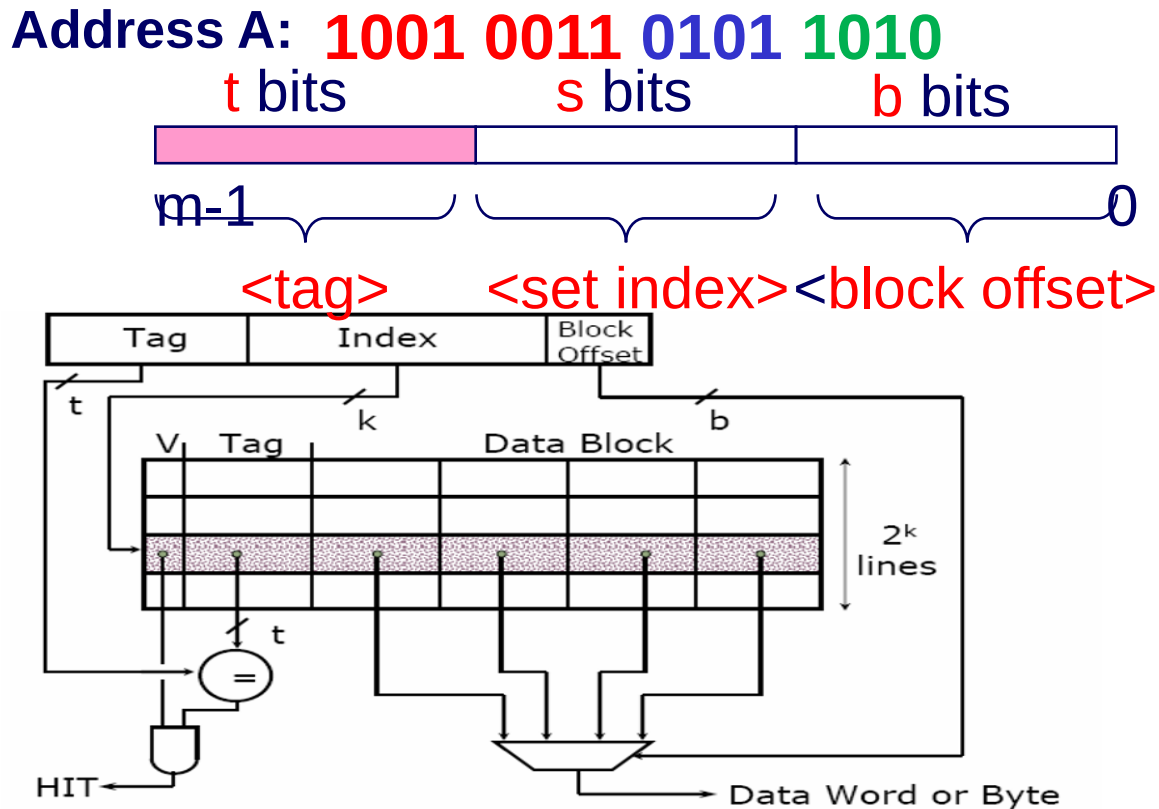
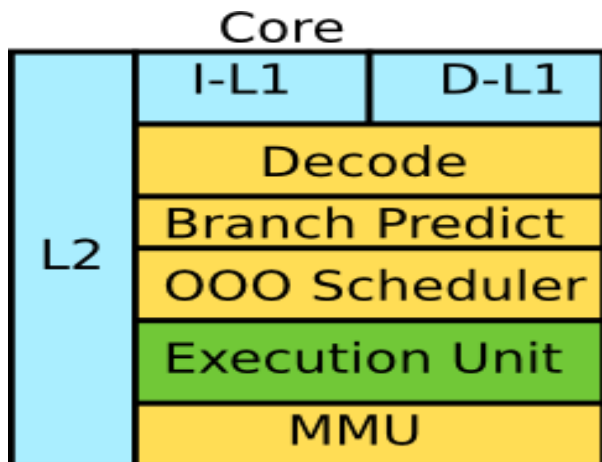


# Routers and Tiles



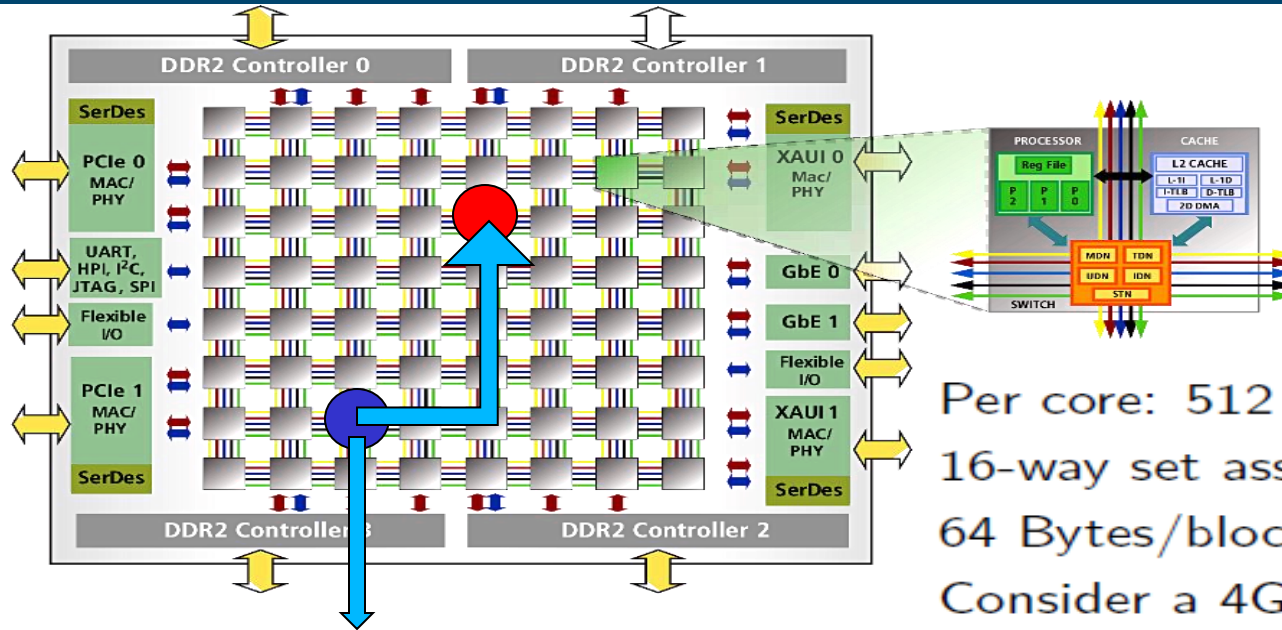
- ❖ East, West, North and South neighbors
- ❖ Packets are divided into flow control units called flits
- ❖ L1 and L2 cache misses create NoC traffic packets

# What is the role of on-chip cache ?





# On-Chip Cache Address Mapping



Per core: 512 KB

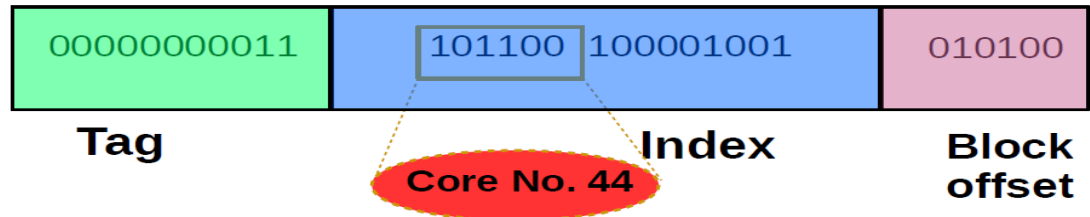
16-way set associative mapping

64 Bytes/block

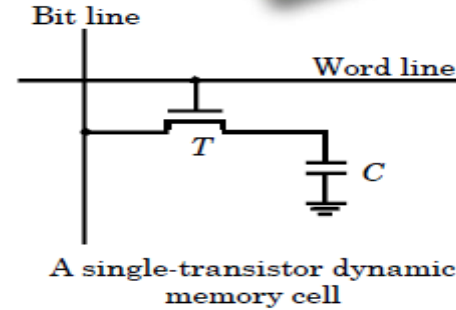
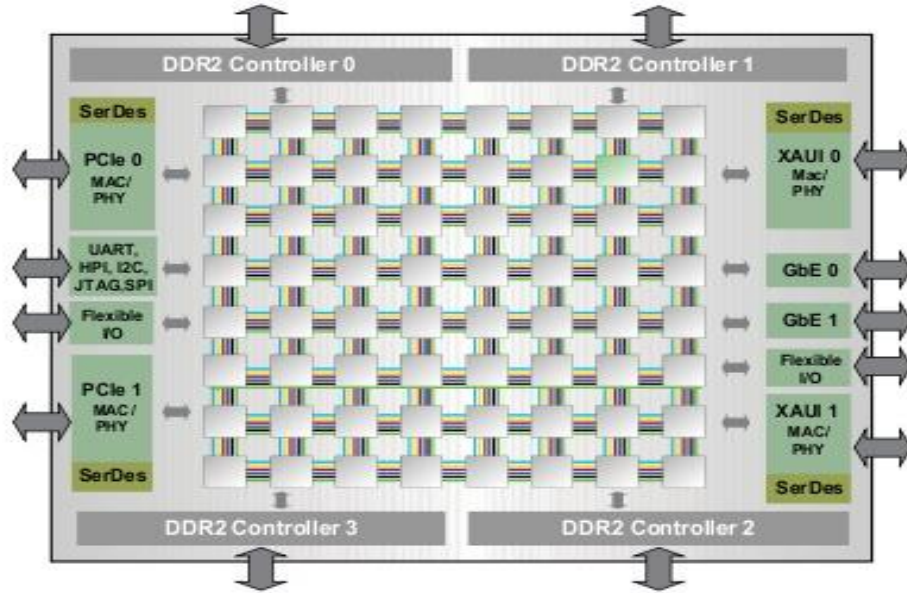
Consider a 4GB RAM (  $2^{32}$  bits)

Hex Address : 0x764254

|         |         |              |
|---------|---------|--------------|
| 11 bits | 15 bits | 6 bits       |
| Tag     | Index   | Block offset |



# What is the role of memory controllers?



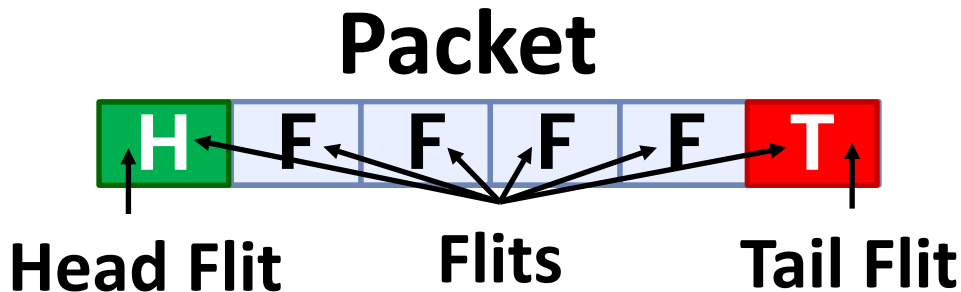
# Packets & Flits

## ❖ Packet

- ❖ Unit of transfer for network

## ❖ Flit

- ❖ Basic unit of transfer between a pair of routers
- ❖ Unit of flow control within network





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